

SmartPoi Controls User Manual

High-level overview of features and user instructions

Note: This manual provides first-person user instructions for operating the SmartPoi Controls web application. It does not cover technical implementation details.

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Getting Started

Initial Setup

1. Ensure your POIs (Main and Aux) are powered on and connected to the same Wi-Fi network as your device.
2. Open the SmartPoi Controls web application in your browser.
3. The application will automatically attempt to discover your POIs on the network.

Interface Overview

The application is organized into seven tabs at the top of the screen:

- **Controls:** Basic POI control and network settings
- **Image Management:** View and manage images on POIs
- **Upload Bin:** Upload and convert images for POI display
- **Text to POI:** Create text-based images and upload to POIs
- **File Lists:** View file lists stored on POIs
- **About Us:** Information about the development partnership
- **Smart Magic Bridge:** Direct ZIP file upload to POIs

Controls Tab

Smart Poi

Controls

Image Management

Upload Bin

Text to POI

File Lists

About Us

Smart Magic Bridge

AP Mode

Poi Controls

Main POI: Offline

Aux POI: Offline

Pattern Selection

Generated

IMG 1-5

IMG 6-10

IMG 10-20

IMG 1-52

On/Off Switch

LEDs OFF

UDP Mode

Synchronization

Sync Poi

Adjustments

Speed (0.5-1800s)

Brightness (20-255)

Network Discovery

WiFi Poi IP Address Discovery (Router Mode)

192.168.1.1

Discover POI

- new Fast Discovery, may need to try a few times to work.

Poi need to be connected to router first (GREEN indicator)

192.168.1.1

Set Main

192.168.1.78

Set Aux

Scanning IP: 0 for poi!

Poi Status

Fetch Settings

Main Poi:

Router: N/A
Password: N/A
Channel: N/A
Pattern: N/A
Pixels: ?

Auxiliary Poi:

Router: N/A
Password: N/A
Channel: N/A
Pattern: N/A
Pixels: ?

DANGER ZONE

Update Router Name and Password:

JackalBlade2.4G

Submit

Switch for Router Mode (submit unchecked to turn off): ☐ Submit

Update WiFi Channel (1-13): 3

Update Channel

Pattern Selection

Choose from nine preset patterns to display on your POIs: 1. **Generated** - Algorithmic pattern 2. **IMG 1-5** - Images 1 through 5 3. **IMG 6-10** - Images 6 through 10 4. **IMG 10-20** - Images 10 through 20 5. **IMG 1-52** - All images (1-52) 6. **On/Off Switch** - Toggle patterns 1-5 every time poi starts 7. **LEDs OFF** - Turn off all LEDs 8. **UDP Mode** - Enable UDP control protocol

How to use: Click any pattern button to immediately send that pattern to both POIs.

Synchronization

- **Sync Poi Button:** Click to synchronize the display timing between Main and Aux POIs.

Adjustments

Use sliders to fine-tune POI behavior: - **Speed Control:** Adjust display speed from 0.5 to 1800 seconds (time between image changes) - **Brightness Control:** Adjust LED brightness from 20 to 255

How to use: Drag the sliders left or right. The tooltip shows the current value.

Network Discovery

Automatically discover POIs on your network using Fast Discovery: 1. Enter your router's IP address (for AP mode choose 192.168.1.1) 2. Click **Discover POI** 3. The system will scan for connected POIs

Note: POIs show a GREEN indicator light in Router Mode, Blue/Red for default Access Point mode **Tip:** Fast Discovery may need to be attempted a few times to work successfully. ### Manual IP Configuration If automatic discovery fails, manually set IP addresses: - **Main POI IP:** Enter IP address and click **Set Main** AP Mode Default: 192.168.1.1 - **Aux POI IP:** Enter IP address and click **Set Aux** AP Mode Default: 192.168.1.78

POI Status Display

View current settings for each POI: - Router name and password (show/hide password option) - WiFi channel (1-13) - Active pattern - Number of pixels detected

How to use: Click **Fetch Settings** to update status information.

Danger Zone

Warning: These settings affect POI network configuration:

1. **Update Router Name and Password:**
 - Enter new router SSID and password
 - Click **Submit** to update POI WiFi credentials
 2. **Router Mode Toggle:**
 - Check/uncheck the box to enable/disable router mode
 - Click **Submit** to apply
 - Note: Poi will now alternate between Router/AP mode on reboot
 3. **Update WiFi Channel:**
 - Enter channel number (1-13)
 - Click **Update Channel**
-

Image Management Tab

Image Management

Fetch and Display All Images

Current px: 60

Fetch Main Poi Images

Fetch Aux Poi Images

WS2812?

Current: CUSTOM (40%)

APA102 | WS2812 | CUSTOM

Main Poi

a.bin	b.bin	c.bin	d.bin
e.bin	f.bin	g.bin	h.bin
i.bin	j.bin	k.bin	l.bin
m.bin	n.bin	o.bin	p.bin
q.bin	r.bin	s.bin	t.bin
u.bin	v.bin	w.bin	x.bin
y.bin	z.bin	A.bin	B.bin
C.bin	D.bin	E.bin	F.bin

Aux Poi

a.bin	b.bin	c.bin	d.bin
e.bin	f.bin	g.bin	h.bin
i.bin	j.bin	k.bin	l.bin
m.bin	n.bin	o.bin	p.bin
q.bin	r.bin	s.bin	t.bin
u.bin	v.bin	w.bin	x.bin
y.bin	z.bin	A.bin	B.bin
C.bin	D.bin	E.bin	F.bin

! DELETE ALL IMAGES ON BOTH POIs !

This will permanently remove all images from both POIs!

Fetching Images

- **Fetch and Display All Images:** Retrieves images from both POIs and displays them in grids
- **Fetch Main Poi Images:** Retrieves images only from the Main POI
- **Fetch Aux Poi Images:** Retrieves images only from the Aux POI

Image Interaction

- **View Enlarged:** Click any image to open it (and display on poi)
- **Long Press/Right-click:** Opens context menu with options:
 - Upload an image to POI
 - Delete image
- **Desktop Browser Drag/Drop:** drag images from file browser onto image placeholder to upload to POI ### Pixel Control
- View current pixel count for each POI (should be fetched from poi automatically)

LED Strip Type

Toggle between APA102, WS2812, or CUSTOM LED strip types.

Note: Custom strip type opens a compression settings dialog (20-80%) for specialized LED setups. Try 40% for WS2812 longer strips - compresses horizontally to fit slow refresh rate strips - default 50% for WS2812, 0% for APA102 ### Delete All Images **Warning:** This action is permanent!
- Click the red ! **DELETE ALL IMAGES ON BOTH POIs !** button

Upload Bin Tab



Controls

Image Management

Upload Bin

Text to POI

File Lists

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Smart Magic Bridge

AP Mode

Upload Pics

Browse...

No files selected.

Upload and convert

LED Strip Type

WS2812?

Current: CUSTOM (40%)

Number of Pixels

Current px: 60

1. Select files 2. Drag to reorder 3. Click upload

Correct number of pixels should be updated from poi.

 **Text to POI**

Text Input

HELLO HOW ARE YOU

Canvas Height (36-256px)
60

Aspect Ratio:
1:1
3:2
2:1
5:2

Text Color

Preview

Text above to

90×60 pixels

Font: Font loaded ✓

Upload to POI

Target POI:
Both POIs
Main POI
Aux POI

File Selection

1. Click "Browse" to select image files (multiple selection supported)
2. Selected files appear in a draggable list

File Reordering

- Drag files up/down in the list to change upload order
- Files are uploaded in the order shown

LED Configuration

- **LED Strip Type:** Toggle between APA102, WS2812, or CUSTOM
- **Number of Pixels:** Adjust pixel count (automatically detected from POI)
- **Custom Compression:** When CUSTOM is selected, a compression settings dialog (20-80%) appears

Upload Process

1. Select files
 2. Reorder if desired
 3. Click **Upload and convert**
 4. Files are converted and uploaded to POI
-

Text to POI Tab

✨ Text to POI

Text Input

HELLO

Canvas Height (36-256px)

60

Aspect Ratio:

1:1

3:2

2:1

5:2

Text Color

Red

Green

Blue

Yellow

Magenta

Cyan

Preview

HELLO

90×60 pixels Font: Font loaded ✓

Upload to POI

Target POI:

Both POIs

Main POI

Aux POI

Connected

Filename:

a.bin

a.bin b.bin c.bin d.bin

Generate Image

Upload to POI

How to Use

1. Enter text in the input field above
2. Adjust height and aspect ratio as needed
3. Select text color from palette or custom picker
4. Click "Generate Image" to preview
5. Select target POI and filename
6. Click "Upload to POI" to send to device

***Note:** The custom font will load automatically. If font loading fails, system font will be used.*

Text Creation

1. **Enter Text:** Type your message in the text input field
2. **Adjust Height:** Set canvas height (choose your actual POI size)
3. **Select Aspect Ratio:** Choose from 1:1, 3:2, 2:1, or 5:2
4. **Choose Color:**
 - Use color picker for custom colors
 - Click color swatches for presets (red, green, blue, yellow, magenta, cyan)

Preview

- Generated text appears on the canvas in real-time
- Canvas dimensions display below preview
- Font loading status shown (custom or system font)

Upload Configuration

1. **Target POI:** Select Both, Main, or Aux POI
2. **Filename:** Choose from suggestions (a.bin, b.bin, c.bin, d.bin) or enter custom name
3. **Connection Status:** Shows if POI is reachable

Upload Process

1. Click **Generate Image** to create preview
2. Verify text appearance
3. Click **Upload to POI** to send to selected device(s)

NOTE:

- Text to POI works in AP Mode only!
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File Lists Tab

Viewing Files

- **Get Main POI Files:** Display all files stored on Main POI
- **Get Aux POI Files:** Display all files stored on Aux POI

File Information

File lists show: - Image files (.bin format) - Configuration files - System files

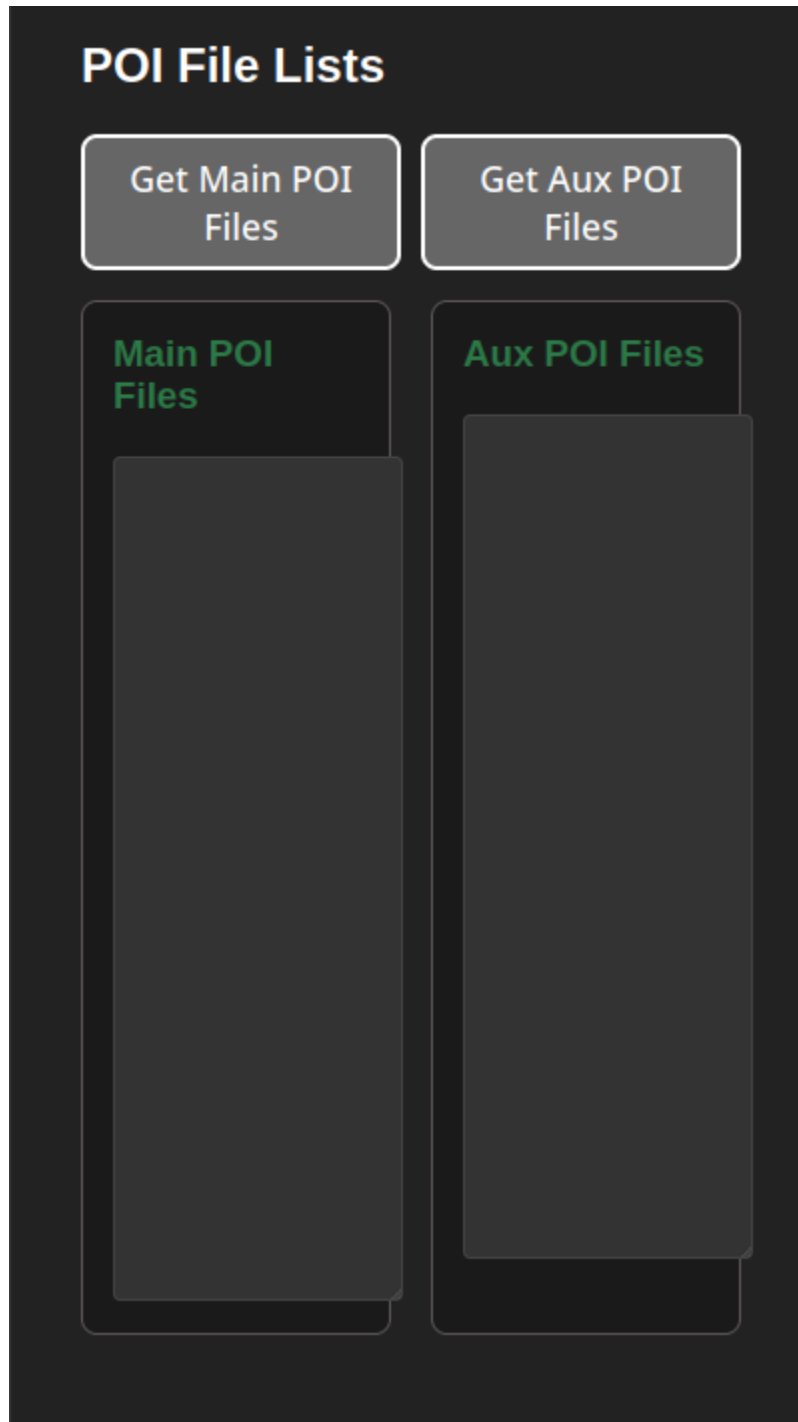


Figure 1: Files1

About Us Tab

Partnership Information

Learn about the collaboration between: - **DevSoft** (South Africa): IoT and embedded systems specialists - **Hoop Roots** (Performance equipment): Hardware design and manufacturing

Contact Information

- DevSoft website: devsoft.co.za
 - Hoop Roots website: hoop-roots.com
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Smart Magic Bridge Tab

Direct ZIP Upload

- ZIP files, create in magicpoi.com (patreon subscribers only)
- **Sign up to Patreon here**
- Upload ZIP files directly to POIs without the need for individual file processing:
 1. Click to select a ZIP file
 2. File contents are listed for verification
 3. Click **Upload to POI**
 4. Progress and status displayed

Clearing Uploads

- Click **Clear** to remove selected ZIP file
-

Upgrading POI firmware (ESP32 version only)

Subscribe to Patreon to receive the latest firmware updates - connect to poi - open web browser - visit <http://192.168.1.78/update> (Auxillary) and - <http://192.168.1.1/update> (Main) - update Auxillary, then Main (choose correct downloaded .bin file) - usually filesystem does not need upgrading (check release notes) - **Router mode: use actual poi IP - Make sure to not upload Auxillary Firmware to Main Poi by mistake!!**

Tips and Best Practices

General

- Keep POIs and control device on the same Wi-Fi network



Our Partnership

Bringing Innovation to Performance

The SmartPoi system is the result of an international collaboration between:

DevSoft

South African software specialists in IoT and embedded systems

Bringing expertise in:

- LED synchronization technology
- Wireless communication systems
- Performance optimization algorithms



[company website](#)

Hoop Roots

equipamentos para performance de Luzes

Contributing:

- Innovative hardware design
- Quality manufacturing
- Testing and sales



[Instagram](#)

Figure 2: About1

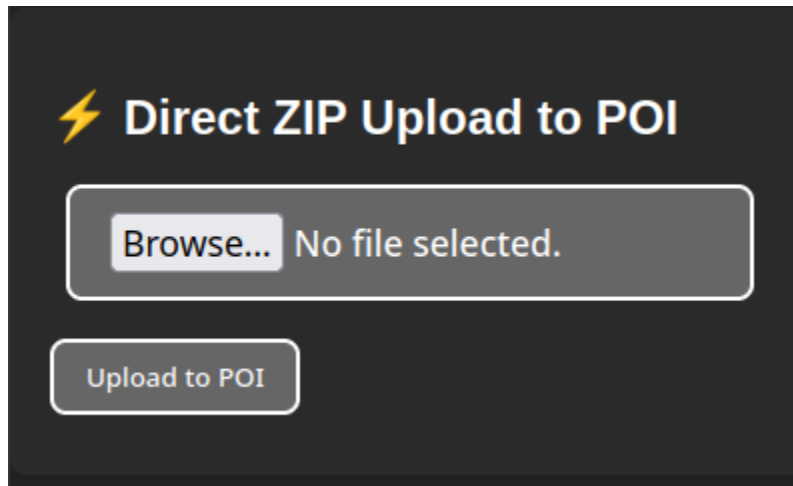


Figure 3: Magic1

- Use the **Fetch Settings** button to verify connection
- Start with lower brightness settings and increase as needed (SmartPoi always start up with Brightness 20 to save battery)

Image Management

- Regularly back up important images
- 1000's of poi images available from magicpoi.com (Patreon subscription required) ### Text Creation
- Shorter text works better with limited pixel displays
- Use high-contrast colors for better visibility
- Test aspect ratios to find optimal text fit

Troubleshooting

- If POIs don't appear, check Wi-Fi connection and indicator lights
- Use manual IP configuration if automatic discovery fails
- Restart POIs if connection issues persist

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